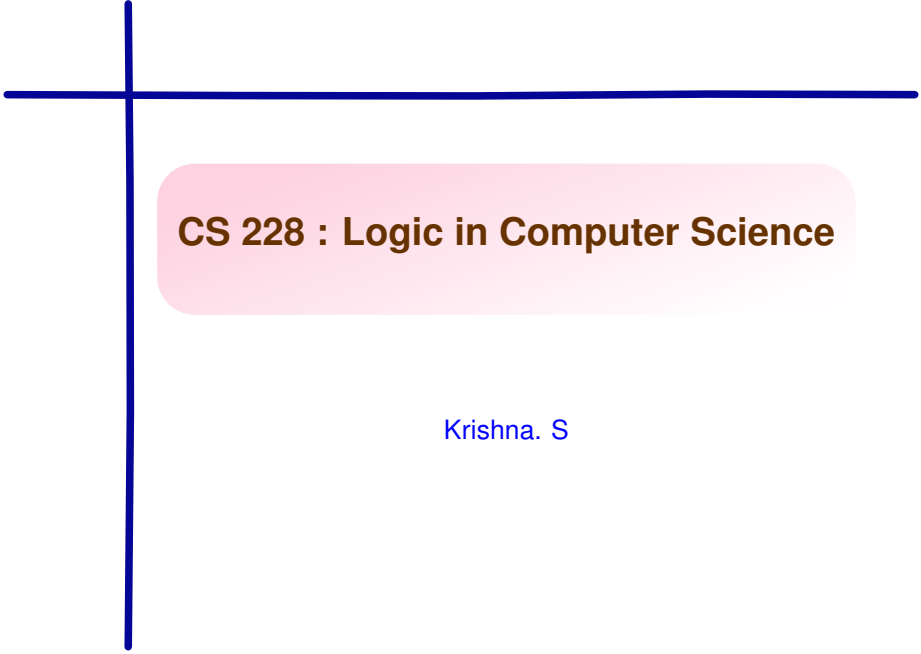




CS 228 : Logic in Computer Science

Krishna. S

First Order Logic

FOL

Extends propositional logic

- ▶ Propositional logic : atomic formulas have no internal structure
- ▶ FOL : atomic formulas are predicates that assert a relationship between certain elements
- ▶ Quantification in FOL : ability to assert that a certain property holds for all elements or only for some element.
- ▶ Formulae in FOL are over some signature.

Signatures

- ▶ A **vocabulary** or **signature** τ is a set consisting of
 - ▶ constants $c_1, c_2, \dots$
 - ▶ Relation symbols $R_1, R_2, \dots$, each with some arity k , denoted R_i^k
 - ▶ Function symbols $f_1, \dots$ each with some arity k , denoted f_i^k
- ▶ We look at finite signatures
- ▶ $\tau = (E^2, F^3, f^1)$ is a finite signature with two relations, E with arity 2 and F with arity 3, and a function f with arity 1

Symbols in First Order Logic

Formulae of FO, over signature τ , are sequences of symbols, where each symbol is one of the following:

Symbols in First Order Logic

Formulae of FO, over signature τ , are sequences of symbols, where each symbol is one of the following:

- ▶ The symbols $\perp, \top$ called **false, true**

Symbols in First Order Logic

Formulae of FO, over signature τ , are sequences of symbols, where each symbol is one of the following:

- ▶ The symbols $\perp, \top$ called **false, true**
- ▶ An element of the infinite set $\mathcal{V} = \{x_1, x_2, \dots\}$ called **variables**

Symbols in First Order Logic

Formulae of FO, over signature τ , are sequences of symbols, where each symbol is one of the following:

- ▶ The symbols $\perp, \top$ called **false, true**
- ▶ An element of the infinite set $\mathcal{V} = \{x_1, x_2, \dots\}$ called **variables**
- ▶ Constants, relations and functions from τ

Symbols in First Order Logic

Formulae of FO, over signature τ , are sequences of symbols, where each symbol is one of the following:

- ▶ The symbols $\perp, \top$ called false, true
- ▶ An element of the infinite set $\mathcal{V} = \{x_1, x_2, \dots\}$ called variables
- ▶ Constants, relations and functions from τ
- ▶ The symbols $\rightarrow, \neg, \wedge, \vee$

Symbols in First Order Logic

Formulae of FO, over signature τ , are sequences of symbols, where each symbol is one of the following:

- ▶ The symbols $\perp, \top$ called **false, true**
- ▶ An element of the infinite set $\mathcal{V} = \{x_1, x_2, \dots\}$ called **variables**
- ▶ Constants, relations and functions from τ
- ▶ The symbols $\rightarrow, \neg, \wedge, \vee$
- ▶ The symbol $\forall$ called the **universal quantifier**

Symbols in First Order Logic

Formulae of FO, over signature τ , are sequences of symbols, where each symbol is one of the following:

- ▶ The symbols $\perp, \top$ called **false, true**
- ▶ An element of the infinite set $\mathcal{V} = \{x_1, x_2, \dots\}$ called **variables**
- ▶ Constants, relations and functions from τ
- ▶ The symbols $\rightarrow, \neg, \wedge, \vee$
- ▶ The symbol $\forall$ called the **universal quantifier**
- ▶ The symbol $\exists$ called the **existential quantifier**

Symbols in First Order Logic

Formulae of FO, over signature τ , are sequences of symbols, where each symbol is one of the following:

- ▶ The symbols $\perp, \top$ called **false, true**
- ▶ An element of the infinite set $\mathcal{V} = \{x_1, x_2, \dots\}$ called **variables**
- ▶ Constants, relations and functions from τ
- ▶ The symbols $\rightarrow, \neg, \wedge, \vee$
- ▶ The symbol $\forall$ called the **universal quantifier**
- ▶ The symbol $\exists$ called the **existential quantifier**
- ▶ The symbols $($ and $)$ called **paranthesis**

Terms

Given a signature τ , the set of τ -terms are defined inductively as follows.

- ▶ Each variable is a term
- ▶ Each constant symbol is a term
- ▶ If $t_1, \dots, t_k$ are terms and f is a k -ary function, then $f(t_1, \dots, t_k)$ is a term
- ▶ Ground Terms : Terms without variables. For instance $f(c_1, \dots, c_k)$ for constants $c_1, \dots, c_k$.

Well formed Formulae

A well-formed formula (wff) over a signature τ is inductively defined as follows:

Well formed Formulae

A well-formed formula (wff) over a signature τ is inductively defined as follows:

- ▶ $\perp, \top$ are wff

Well formed Formulae

A well-formed formula (wff) over a signature τ is inductively defined as follows:

- ▶ $\perp, \top$ are wff
- ▶ If t_1, t_2 are terms, then $t_1 = t_2$ is a wff

Well formed Formulae

A well-formed formula (wff) over a signature τ is inductively defined as follows:

- ▶ $\perp, \top$ are wff
- ▶ If t_1, t_2 are terms, then $t_1 = t_2$ is a wff
- ▶ If t_i is a term, for $1 \leq i \leq k$ and R is a k -ary relation symbol in τ , then $R(t_1, \dots, t_k)$ is a wff

Well formed Formulae

A well-formed formula (wff) over a signature τ is inductively defined as follows:

- ▶ $\perp, \top$ are wff
- ▶ If t_1, t_2 are terms, then $t_1 = t_2$ is a wff
- ▶ If t_i is a term, for $1 \leq i \leq k$ and R is a k -ary relation symbol in τ , then $R(t_1, \dots, t_k)$ is a wff
- ▶ If φ and ψ are wff, then $\varphi \rightarrow \psi, \varphi \wedge \psi, \varphi \vee \psi, \neg \psi$ are all wff

Well formed Formulae

A well-formed formula (wff) over a signature τ is inductively defined as follows:

- ▶ $\perp, \top$ are wff
- ▶ If t_1, t_2 are terms, then $t_1 = t_2$ is a wff
- ▶ If t_i is a term, for $1 \leq i \leq k$ and R is a k -ary relation symbol in τ , then $R(t_1, \dots, t_k)$ is a wff
- ▶ If φ and ψ are wff, then $\varphi \rightarrow \psi, \varphi \wedge \psi, \varphi \vee \psi, \neg \psi$ are all wff
- ▶ If φ is a wff and x is a variable, then $(\forall x)\varphi$ and $(\exists x)\varphi$ are wff

Well formed Formulae

A well-formed formula (wff) over a signature τ is inductively defined as follows:

- ▶ $\perp, \top$ are wff
- ▶ If t_1, t_2 are terms, then $t_1 = t_2$ is a wff
- ▶ If t_i is a term, for $1 \leq i \leq k$ and R is a k -ary relation symbol in τ , then $R(t_1, \dots, t_k)$ is a wff
- ▶ If φ and ψ are wff, then $\varphi \rightarrow \psi, \varphi \wedge \psi, \varphi \vee \psi, \neg \psi$ are all wff
- ▶ If φ is a wff and x is a variable, then $(\forall x)\varphi$ and $(\exists x)\varphi$ are wff
- ▶ The second and third are **atomic** formulae.
- ▶ If a wff F occurs as part of another wff G , then F is called a **sub formula** of G .

Logical Abbreviations : Boolean Connectives

- ▶ $\neg\varphi = \varphi \rightarrow \perp$
- ▶ $\top = \neg\perp$
- ▶ $\varphi \vee \psi = \neg\varphi \rightarrow \psi$
- ▶ $\varphi \wedge \psi = \neg(\neg\varphi \vee \neg\psi)$
- ▶ $\exists x.\varphi = \neg(\forall x.\neg\varphi)$
- ▶ Precedence of operators : Quantifiers and negation highest, followed by $\vee$, $\wedge$, followed by $\rightarrow$.
 - ▶ $\forall x P(x) \wedge R(x)$ is $[\forall x.[P(x)]] \wedge R(x)$

An Example

Consider the signature $\tau = \{R\}$ where R is a binary relation. The following are FO formulae over this signature.

- ▶ $\forall x R(x, x)$ Reflexivity
- ▶ $\forall x (R(x, x) \rightarrow \perp)$ Irreflexivity
- ▶ $\forall x \forall y (R(x, y) \rightarrow R(y, x))$ Symmetry
- ▶ $\forall x \forall y \forall z (R(x, y) \rightarrow (R(y, z) \rightarrow R(x, z)))$ Transitivity

First-Order Logic : Semantics

Structures

- ▶ A structure $\mathcal{A}$ of signature τ consists of

Structures

- ▶ A structure $\mathcal{A}$ of signature τ consists of
 - ▶ A non-empty set A or $u(\mathcal{A})$ called the **universe**

Structures

- ▶ A structure $\mathcal{A}$ of signature τ consists of
 - ▶ A non-empty set A or $u(\mathcal{A})$ called the **universe**
 - ▶ For each constant c in the signature τ , a fixed element $c_{\mathcal{A}}$ is assigned from the universe A

Structures

- ▶ A structure $\mathcal{A}$ of signature τ consists of
 - ▶ A non-empty set A or $u(\mathcal{A})$ called the **universe**
 - ▶ For each constant c in the signature τ , a fixed element $c_{\mathcal{A}}$ is assigned from the universe A
 - ▶ For each k -ary relation R^k in the signature τ , a set of k -tuples from A^k is assigned to $R^{\mathcal{A}}$

Structures

- ▶ A structure $\mathcal{A}$ of signature τ consists of
 - ▶ A non-empty set A or $u(\mathcal{A})$ called the **universe**
 - ▶ For each constant c in the signature τ , a fixed element $c_{\mathcal{A}}$ is assigned from the universe A
 - ▶ For each k -ary relation R^k in the signature τ , a set of k -tuples from A^k is assigned to $R^{\mathcal{A}}$
 - ▶ The structure $\mathcal{A}$ is finite if A (or $u(\mathcal{A})$) is finite

Examples of Structures : A Graph

- ▶ $\tau = \{E\}$, with E binary.

Examples of Structures : A Graph

- ▶ $\tau = \{E\}$, with E binary.
 - ▶ A graph structure over τ is $\mathcal{G} = (V, E^{\mathcal{G}})$,
 - ▶ The **universe** $u(\mathcal{G})$ is the set of vertices V
 - ▶ The relation E is the edge relation

Examples of Structures : A Graph

- ▶ $\tau = \{E\}$, with E binary.
 - ▶ A graph structure over τ is $\mathcal{G} = (V, E^{\mathcal{G}})$,
 - ▶ The **universe** $u(\mathcal{G})$ is the set of vertices V
 - ▶ The relation E is the edge relation
 - ▶ $\mathcal{G} = (V = \{1, 2, 3, 4\}, E^{\mathcal{G}} = \{(1, 2), (2, 3), (3, 4), (1, 1)\})$. We could just as well draw the graph for convenience.

Examples of Structures : An Order

- ▶ $\tau = \{<, S\}$ with $<, S$ binary.

Examples of Structures : An Order

- ▶ $\tau = \{<, S\}$ with $<$, S binary.
 - ▶ A finite order structure over τ is $\mathcal{O} = (O, <^{\mathcal{O}}, S^{\mathcal{O}})$
 - ▶ The universe $u(\mathcal{O})$ is the finite ordered set O
 - ▶ $<^{\mathcal{O}}$ is the ordering on O and $S^{\mathcal{O}}$ is the successor on O

Examples of Structures : An Order

- ▶ $\tau = \{<, S\}$ with $<, S$ binary.
 - ▶ A finite order structure over τ is $\mathcal{O} = (O, <^{\mathcal{O}}, S^{\mathcal{O}})$
 - ▶ The universe $u(\mathcal{O})$ is the finite ordered set O
 - ▶ $<^{\mathcal{O}}$ is the ordering on O and $S^{\mathcal{O}}$ is the successor on O
 - ▶ $\mathcal{O} = (O = \{1, 2, 4\}, <^{\mathcal{O}} = \{(1, 2), (1, 4), (2, 4)\}, S^{\mathcal{O}} = \{(1, 2)\})$

Examples of Structures : A Word

- ▶ $\tau = \{<, S, Q_a, Q_b\}$, where $<, S$ are binary, Q_a, Q_b are unary relations.

Examples of Structures : A Word

- ▶ $\tau = \{<, S, Q_a, Q_b\}$, where $<, S$ are binary, Q_a, Q_b are unary relations.
 - ▶ A word structure $\mathcal{W} = (u(\mathcal{W}), <^{\mathcal{W}}, S^{\mathcal{W}}, Q_a^{\mathcal{W}}, Q_b^{\mathcal{W}})$

Examples of Structures : A Word

- ▶ $\tau = \{<, S, Q_a, Q_b\}$, where $<, S$ are binary, Q_a, Q_b are unary relations.
 - ▶ A word structure $\mathcal{W} = (u(\mathcal{W}), <^{\mathcal{W}}, S^{\mathcal{W}}, Q_a^{\mathcal{W}}, Q_b^{\mathcal{W}})$
 - ▶ The universe $u(\mathcal{W})$ consists of the positions in a word W over symbols a, b

Examples of Structures : A Word

- ▶ $\tau = \{<, S, Q_a, Q_b\}$, where $<, S$ are binary, Q_a, Q_b are unary relations.
 - ▶ A word structure $\mathcal{W} = (u(\mathcal{W}), <^{\mathcal{W}}, S^{\mathcal{W}}, Q_a^{\mathcal{W}}, Q_b^{\mathcal{W}})$
 - ▶ The universe $u(\mathcal{W})$ consists of the positions in a word W over symbols a, b
 - ▶ $<^{\mathcal{W}}$ is the ordering relation on the positions of W

Examples of Structures : A Word

- ▶ $\tau = \{<, S, Q_a, Q_b\}$, where $<, S$ are binary, Q_a, Q_b are unary relations.
 - ▶ A word structure $\mathcal{W} = (u(\mathcal{W}), <^{\mathcal{W}}, S^{\mathcal{W}}, Q_a^{\mathcal{W}}, Q_b^{\mathcal{W}})$
 - ▶ The universe $u(\mathcal{W})$ consists of the positions in a word W over symbols a, b
 - ▶ $<^{\mathcal{W}}$ is the ordering relation on the positions of W
 - ▶ $S^{\mathcal{W}}$ is the successor relation on the positions of W

Examples of Structures : A Word

- ▶ $\tau = \{<, S, Q_a, Q_b\}$, where $<, S$ are binary, Q_a, Q_b are unary relations.
 - ▶ A word structure $\mathcal{W} = (u(\mathcal{W}), <^{\mathcal{W}}, S^{\mathcal{W}}, Q_a^{\mathcal{W}}, Q_b^{\mathcal{W}})$
 - ▶ The universe $u(\mathcal{W})$ consists of the positions in a word W over symbols a, b
 - ▶ $<^{\mathcal{W}}$ is the ordering relation on the positions of W
 - ▶ $S^{\mathcal{W}}$ is the successor relation on the positions of W
 - ▶ $Q_a^{\mathcal{W}}$ is the set of positions labeled a in W

Examples of Structures : A Word

- ▶ $\tau = \{<, S, Q_a, Q_b\}$, where $<, S$ are binary, Q_a, Q_b are unary relations.
 - ▶ A word structure $\mathcal{W} = (u(\mathcal{W}), <^{\mathcal{W}}, S^{\mathcal{W}}, Q_a^{\mathcal{W}}, Q_b^{\mathcal{W}})$
 - ▶ The universe $u(\mathcal{W})$ consists of the positions in a word W over symbols a, b
 - ▶ $<^{\mathcal{W}}$ is the ordering relation on the positions of W
 - ▶ $S^{\mathcal{W}}$ is the successor relation on the positions of W
 - ▶ $Q_a^{\mathcal{W}}$ is the set of positions labeled a in W
 - ▶ $Q_b^{\mathcal{W}}$ is the set of positions labeled b in W

Examples of Structures : A Word

- ▶ $\tau = \{<, S, Q_a, Q_b\}$, where $<, S$ are binary, Q_a, Q_b are unary relations.
 - ▶ A word structure $\mathcal{W} = (u(\mathcal{W}), <^{\mathcal{W}}, S^{\mathcal{W}}, Q_a^{\mathcal{W}}, Q_b^{\mathcal{W}})$
 - ▶ The universe $u(\mathcal{W})$ consists of the positions in a word W over symbols a, b
 - ▶ $<^{\mathcal{W}}$ is the ordering relation on the positions of W
 - ▶ $S^{\mathcal{W}}$ is the successor relation on the positions of W
 - ▶ $Q_a^{\mathcal{W}}$ is the set of positions labeled a in W
 - ▶ $Q_b^{\mathcal{W}}$ is the set of positions labeled b in W
 - ▶ The structure with $u(\mathcal{W}) = \{0, 1, 2, \dots, 8\}$,
 $Q_a^{\mathcal{W}} = \{0, 1, 4, 6, 8\}$, $Q_b^{\mathcal{W}} = \{2, 3, 5, 7\}$,
 - ▶ $<^{\mathcal{W}} = \{(0, 1), (0, 2), \dots, (7, 8)\}$, $S^{\mathcal{W}} = \{(0, 1), (1, 2), \dots, (7, 8)\}$
uniquely defines the word $W = aabbababa$.
 - ▶ For convenience, we can just write the word instead of the structure.